Péter Juhász

PhD Researcher in Mathematics

CONTACT

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LinkedIn

shepherd92

EDUCATION

2022 – 2025: PhD in Mathematics
Aarhus University, Denmark

2019 – 2021: MBA in Finance
Budapest University of Technology and Economics, Hungary

2013 – 2016: MSc in Physics
Budapest University of Technology and Economics, Hungary

2013 – 2016: BSc in Physics
Budapest University of Technology and Economics, Hungary

SKILLS

Stochastic Models · Probability · Data Analysis · Python · C++ · SQL · Network Science

LANGUAGES

Hungarian

English

German

CERTIFICATIONS

2021: Software Architect ISAQB

2021: Software Designer Bosch

PROFILE

I have recently finished my PhD in Mathematics at Aarhus University. My research combines stochastic modeling of high-dimensional random systems, probabilistic analysis, and large-scale C++/Python simulations. I am particularly interested in applying mathematical models to real-world problems such as forecasting, risk modeling, and optimization in energy and financial markets. Previously, I worked at Bosch as a machine learning researcher and software architect in automated driving, focusing on environment prediction, data-driven decision-making, and scalable system design.

PROFESSIONAL EXPERIENCE

PhD Researcher in Mathematics — **2022 – 2025**
Aarhus University, Denmark

- Developed **stochastic models** for large-scale, high-dimensional **random systems.**
- Designed **efficient Monte Carlo algorithms in C++ and Python** for hypothesis testing of stochastic network models.
- Proved **theoretical results in stochastic point processes and random graphs.**
- Presented results at several **international conferences on stochastic models**, probability, and data analysis.

Machine Learning Researcher & Software Architect — **2018 – 2022**
Bosch Group, Hungary

- **Patented** a Gaussian regression based **stochastic trajectory prediction model** improving predictive accuracy by $3\times$ and uncertainty estimation by $10\times$ over baseline.
- **Led a team of 11** developing environment prediction models for automated driving as a technical lead.
- **Created a framework for automated ground-truth generation** enabling large-scale data-driven model evaluation.

Software Engineer — **2016 – 2018**
Ericsson Telecommunications, Hungary

- Designed **response time-based load-balancing algorithms** for distributed systems, improving efficiency though quantitative optimization.

AWARDS, GRANTS

2022: PhD Fellowship
Danish Data Science Academy
1 890 000 DKK

2021: Excellence Award
Hungarian National Bank
350 000 HUF

2025: Individual Travel Grant
European Mathematical Society
500 EUR

2024: Academic Mobility Grant
Aarhus University
20 000 DKK

SELECTED PUBLICATIONS, PATENTS

- C. Hirsch, B. Jahnel, and P. Juhász.
  Functional limit theorems for edge counts in dynamic random connection hypergraphs,
  arXiv preprint (2025), arxiv:2507.16270
- P. Juhász.
  Method and device for predicting the trajectory of a traffic participant, and sensor system,
  CNIPA: CN113988353A, EPO: EP3916697A1
  JPO: JP2021190119A USPTO: US2021366274A1 (2021)
- P. Juhász.
  Information propagation in stochastic networks,
  Phys. A (577) (2021), 126070
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay.
  A Bayesian approach to identify Bitcoin users,
  PLoS One (13) (2018), e0207000

SELECTED TEACHING

Advanced Probability Theory — **2025**
10 ECTS Exercise Class

Data Project — **2024**
10 ECTS Computer Project Class

Ordinary Differential Equations — **2023**
5 ECTS Exercise Class